Operating instructions

V 1.0 English

QiTech Aquapath

QiTech Industries GmbH
Landwehrstraße 55
64293 Darmstadt

EC Declaration of Conformity

According to 2006/42/EC, Annex II, No. 1 A

The content of the EC Declaration of Conformity corresponds to Annex II, No. 1 A of Directive 2006/42/EC.

Manufacturer of the machine:

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Representative of the manufacturer:

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Person authorised to compile the technical file:

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General designation	Filament water cooling system
Type	Filament water cooling system
Function	Water cooling of extruded filaments
Model	QiTech Aquapath
Trade name	QiTech Aquapath
Serial number	

We hereby declare that the above-mentioned machine is in conformity with all relevant provisions of the EC Machinery Directive 2006/42/EC. The machine is also in compliance with all relevant provisions of the following EC Directives:

- 2014/30/EU Electromagnetic Compatibility / European EMC Directive
- 2014/35/EU Low Voltage Directive (protection objectives only)
- 2011/65/EU ROHS Directives (ROHS 2)

The following harmonized standards (or parts of these standards) have been applied:

- DIN EN ISO 12100
- DIN EN ISO 20607

Darmstadt, 08.09.2026



Milan von dem Bussche

Read before commissioning!

These operating instructions are intended to enable all people working on the machine to carry out their work safely and correctly. Every person who operates, maintains or repairs the machine must have read the operating instructions, particularly the safety regulations, before commissioning. These operating instructions must always be available on the machine. The operator must ensure that all information necessary for the safe operation of the machine is made available to the operating personnel. If the operating instructions are available in electronic form, the operator must ensure that the operating personnel are able to view the operating instructions. The original version of these operating instructions is written in English. We ask you to read through the relevant instructions carefully and to observe them thoroughly. After unpacking the machine, check for any damage that may have occurred during transport.

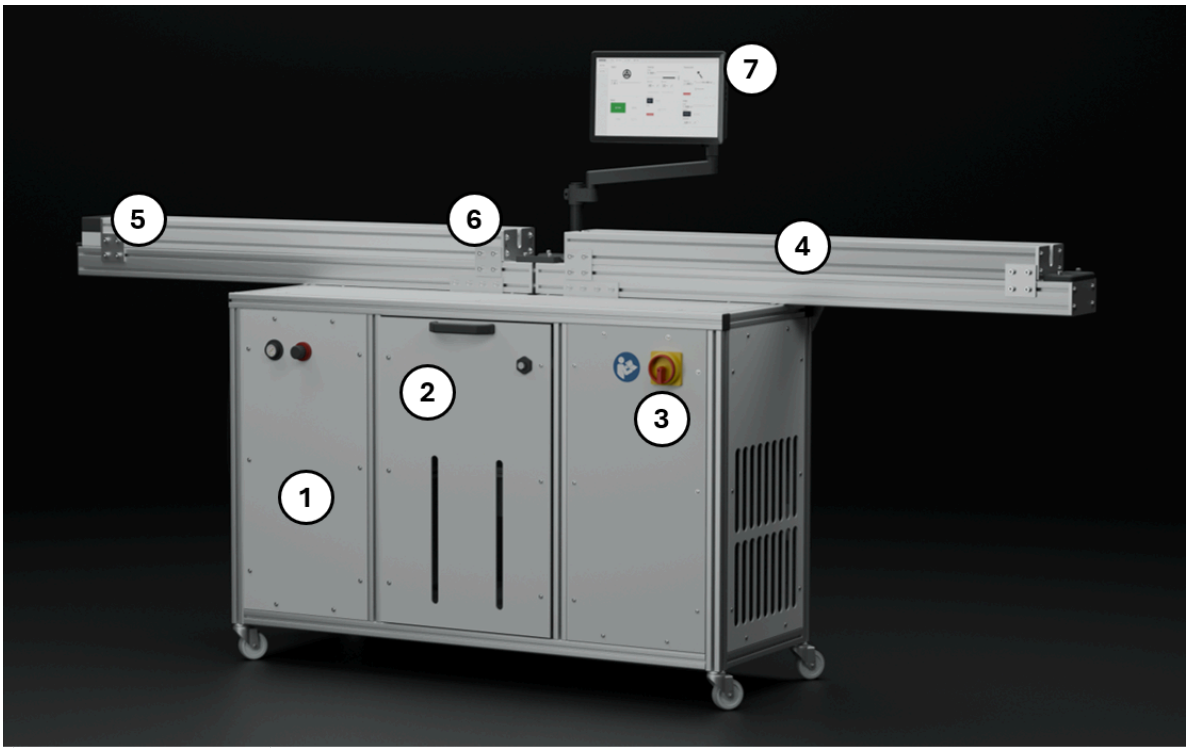
NOTICE: Before using the machine for the first time, please inform yourself about the national regulations (e.g. occupational safety, environmental restrictions). Reproduction and reprinting in any form, even in extracts, requires the written authorization of QiTech Industries GmbH.

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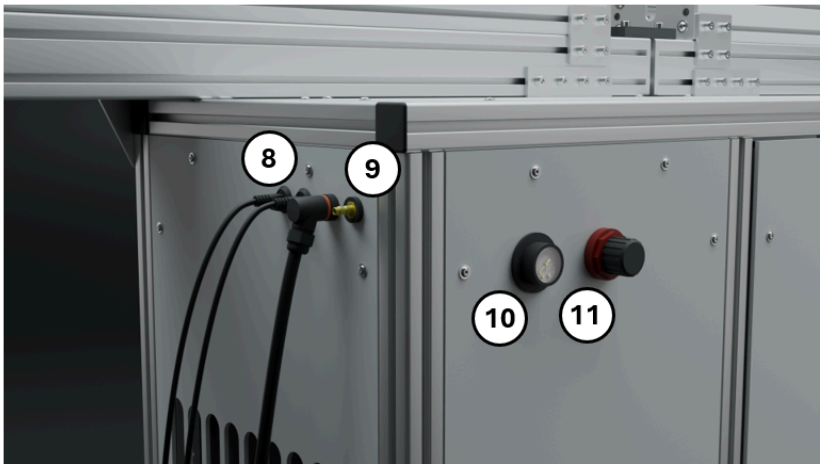
1. Machine description

1.1 Component overview









Item number	Description
1	underframe with heavy-duty casters with levelling feet
2	maintenance door to access the internal components
3	main switch with emergency switching-off function
4	water reservoir with reservoir cover

5	blow-off device
6	guide coins
7	control display
8	connector ports (power supply and LAN)
9	compressed air connection
10	pressure gauge
11	pressure regulator
12	power supply cable (Lapp EPIC M12 5pin)
13	LAN-cable

The following components are located inside the machine and are accessible through the maintenance door:

Each of the two water circuits contains a water pump, a water tank, a water filter, a heating element, a cooling element, a flow sensor and a shut-off valve, which is fitted in the hose downstream of the pump. The pumps convey the water from the tanks into the reservoirs and back. The heating and cooling elements regulate the water of each circuit to its set target temperature, and the flow sensors monitor the water flow through the circuit.

The electrical control cabinet is separated structurally from the water-carrying components.

The design of the main control element depends on how the QiTech Aquapath is operated.

In standalone operation, the QiTech Aquapath is equipped with a red/yellow rotary main switch, which serves as both the main switch and the emergency switching-off device for the machine.

When the QiTech Aquapath is operated as part of a process line together with the QiTech Extruder and the QiTech Winder, a central emergency stop is provided at the cooling unit for easy access from the front of the process line. It switches off the entire process line and needs to be released manually before the process line can be restarted. The main switch on the QiTech Aquapath retains the same function, but it no longer serves as an emergency switching-off device. It functions solely as the main switch of the machine and switches off the QiTech Aquapath and the QiTech Winder due to the power connection system.

The arrangement of the machines and control elements in the process line is shown in the rendering in Annex A.

1.2 Technical specifications

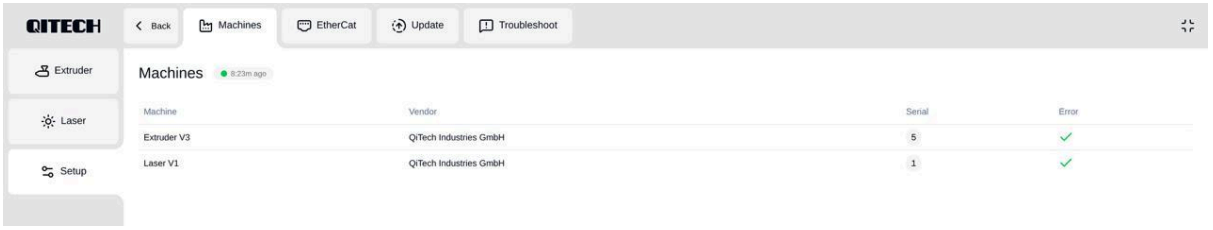
Dimensions and weight	
Length x Width x Height	2870mm x 600mm x 1120mm
Weight	130 kg
Operating weight	162 kg
Electrical specifications	
Rated voltage	230 V
Rated frequency	50 Hz
Rated apparent power	0,759 kVA
Rated current	3,3 A
Process parameters	
Permissible filament diameter	1 – 3 mm
Permissible filament temperature	300 °C
Number of water reservoirs	2
Water tank capacity	20 l, filled 80%
Permissible water temperature range	room temperature - 80 °C
Maximum pressure of compressed air	8 bar
Noise emissions	
Workplace-related emission value (L_pA)	< 70 dB

1.3 Controls and interface

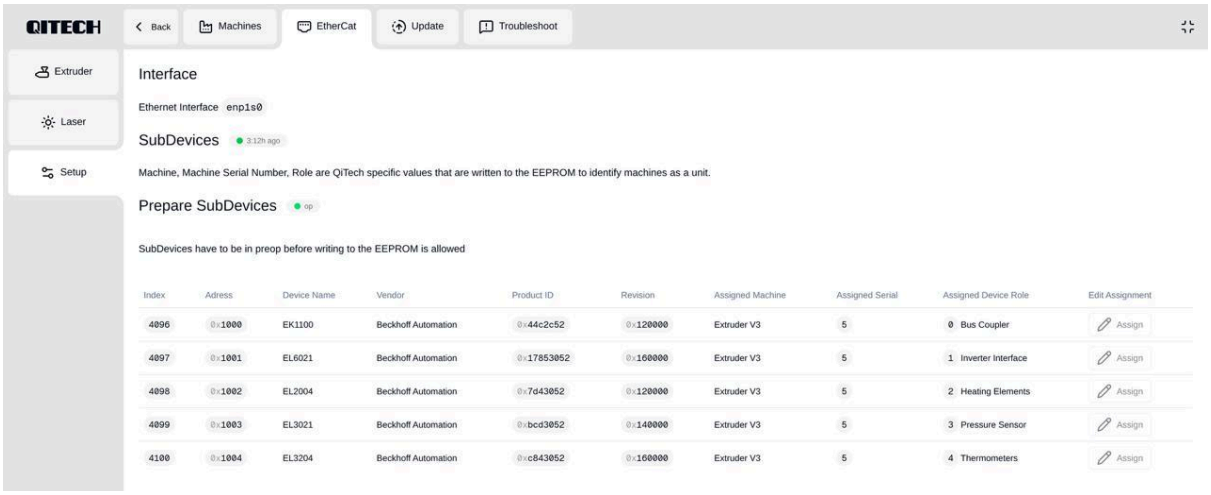
1.3.1 General controls and interface

The QiTech Aquapath is operated entirely via the QiTech Control software. The machine is accessible on the left-hand sidebar, which makes it possible to control multiple machines simultaneously.

The sidebar on the left switches between the views of the connected machine. *Set Up* provides access to machine identification, EtherCAT device configuration, software updates and system troubleshooting functions.

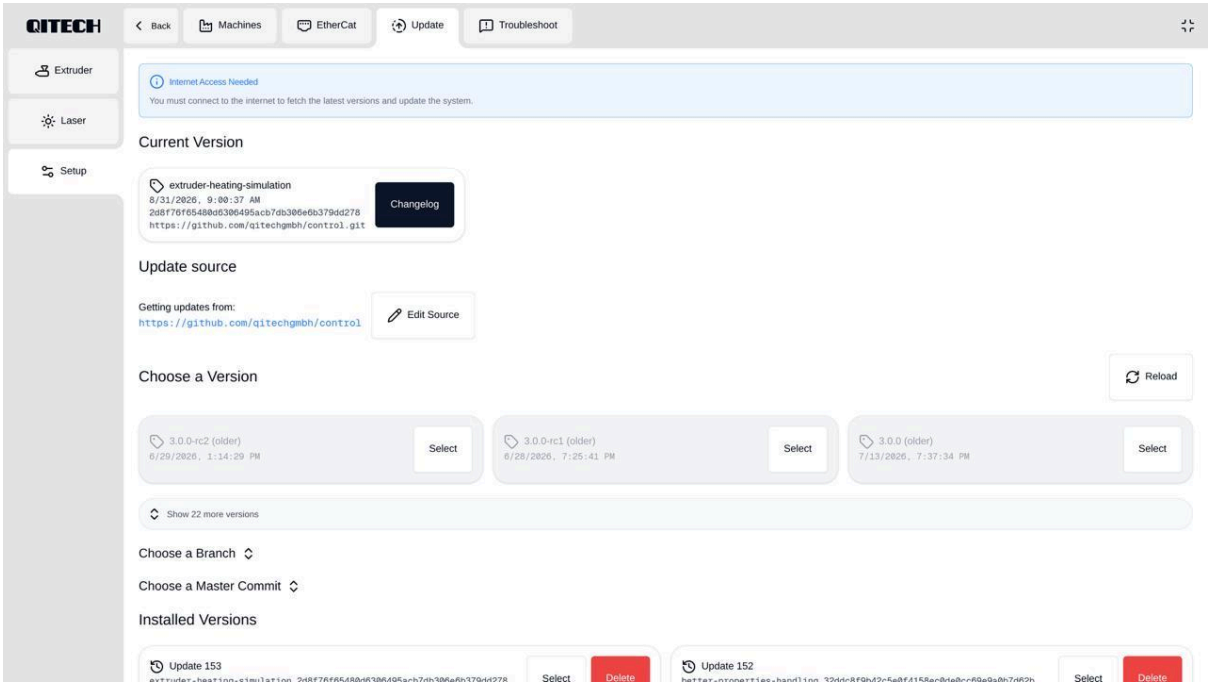


Machines	
Machine	Shows the name and version of the configured machine
Vendor	Shows the manufacturer of the respective machine
Serial	Displays the assigned machine serial number
Error Status	Indicates whether an error is currently detected for the machine



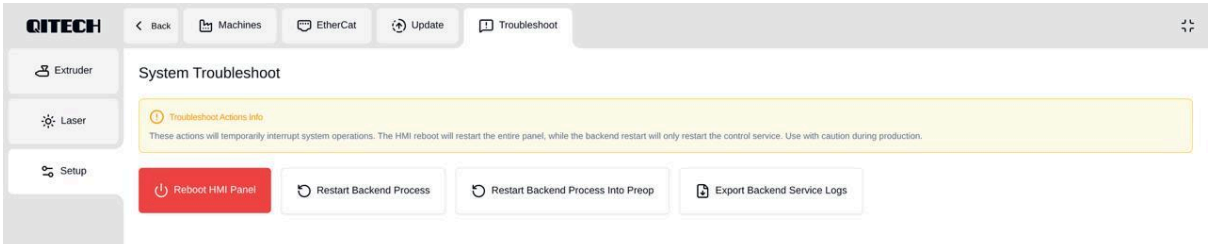
Ethercat	
Ethernet Interface	Shows the Ethernet interface currently used for EtherCAT communication
SubDevices	Displays all detected EtherCAT subdevices connected to the system
Index	Shows the internal index of the respective EtherCAT subdevice
Address	Shows the EtherCAT address assigned to the subdevice
Device Name	Displays the detected EtherCAT device type
Vendor	Shows the manufacturer of the EtherCAT device
Product ID	Displays the EtherCAT product identification number
Revision	Displays the hardware/device revision
Assigned Machine	Shows the machine to which the subdevice is assigned

Assigned Serial	Shows the serial number of the assigned machine
Assigned Device Role	Shows the function of the subdevice within the machine
Edit Assignment	Opens the assignment settings for the selected subdevice



Update	
Current Version	Displays the software version currently installed and running
Changelog	Opens the changelog of the current software version
Update Source	Shows the repository or source from which software updates are retrieved; Edit source if software update source must be changed
Choose a Version	Displays available software versions that can be installed; Select the corresponding software version for installation

Reload	Refreshes the list of available software versions
Choose a Branch	Allows a software branch to be selected manually
Choose a Master Commit	Allows a specific master commit to be selected manually
Select Installed Version	Displays software versions already stored on the system; Select or delete a previously installed software version

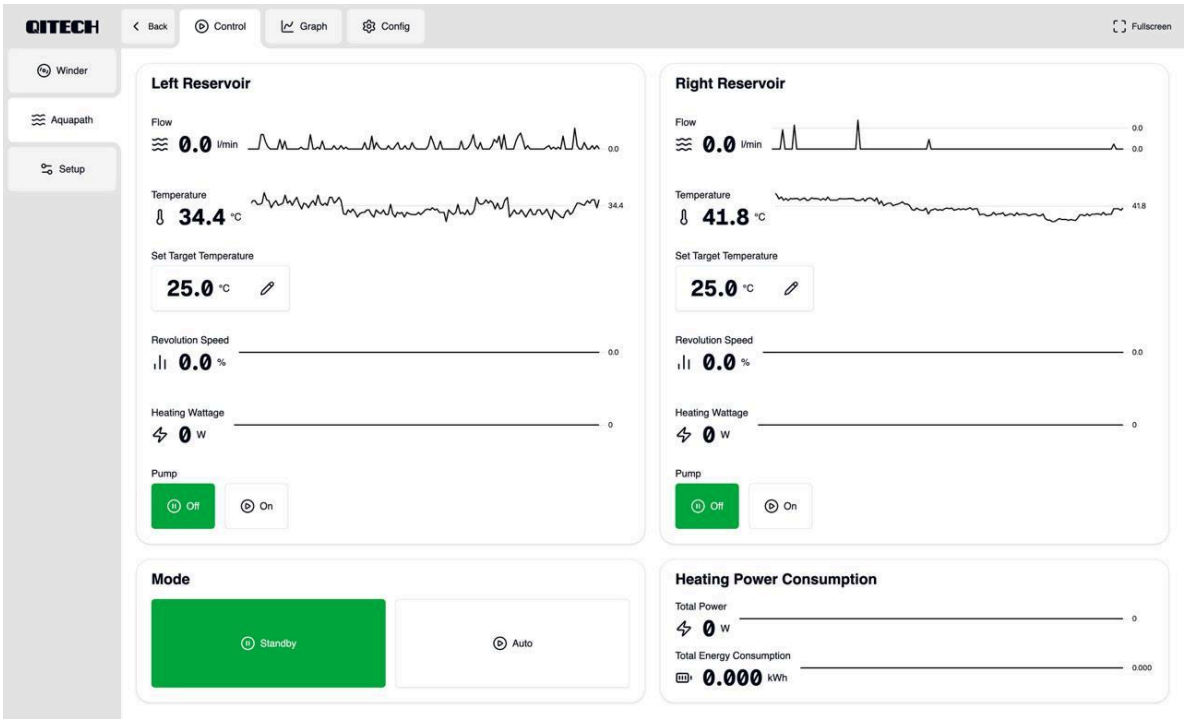


Troubleshoot	
Reboot HMI Panel	Reboots the complete HMI panel and restarts the user interface
Restart Backend Process	Restarts the backend control process without rebooting the complete HMI panel
Restart Backend Process Into PREOP	Restarts the backend process and places the EtherCAT system into PREOP state
Export Backend Service Logs	Exports backend service log files for troubleshooting and diagnostic purposes

1.3.2 Machine-specific controls and interface

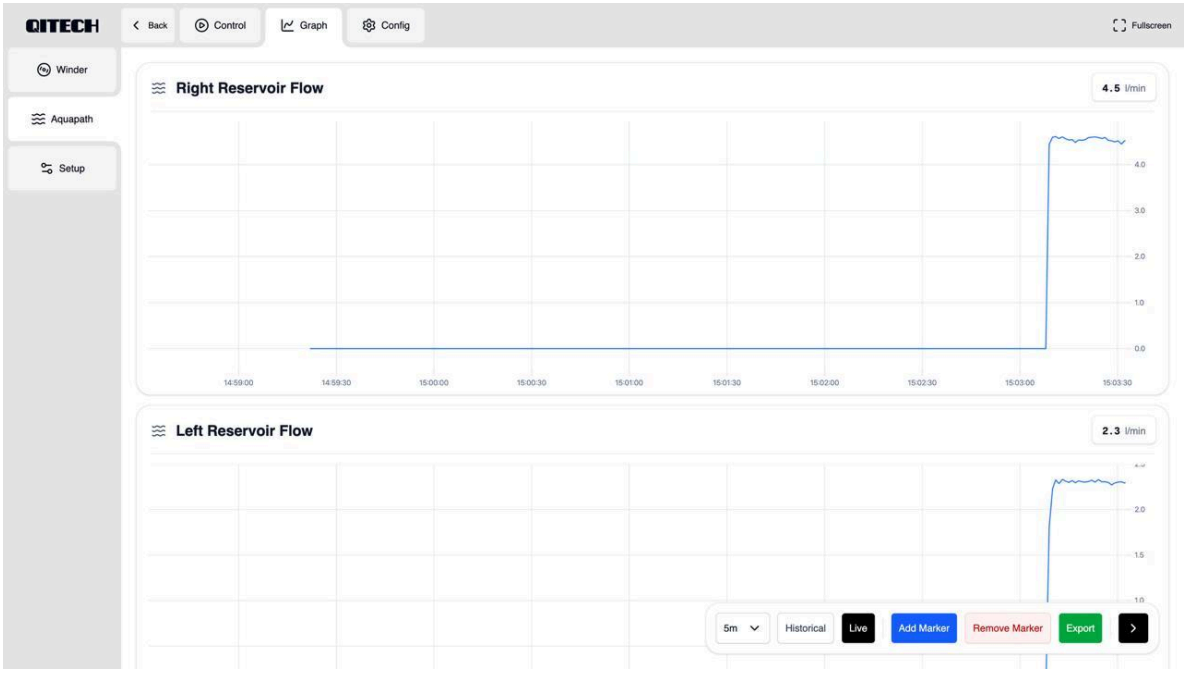
The navigation bar at the top of the screen provides access to the main areas of the QiTech Control software. The *Control* view is the primary operating view, displaying all real-time machine parameters. The *Graphs* view shows process data graphically over time and allows data to be exported in Excel format. The *Config* view provides access to the machine configuration settings. The *Manual* view contains this document, so that all instructions remain available directly at the machine. The *Presets* view is used to save the current

parameter set and to load previously stored ones, which allows a proven set of settings to be recalled for a known material. The *Back* button returns to the machine selection screen, and *Fullscreen* expands the software to the full display area.



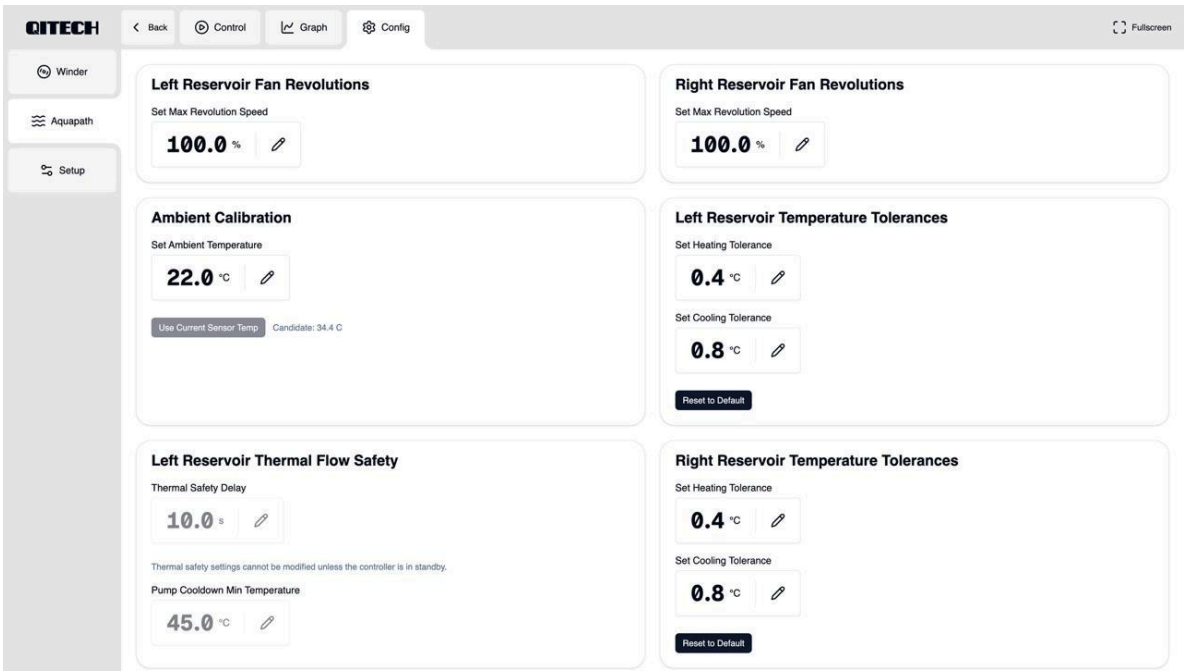
Control	
Left / Right Reservoir	
Flow	Displays the current water flow rate through the circuit in l/min, measured by the flow sensor
Temperature	Displays the current water temperature of the reservoir in °C
Set Target Temperature	Sets the target temperature of the reservoir; adjustable via the edit button. A status indicator shows when the circuit is actively cooling or heating to reach the target temperature
Revolution Speed	Displays the current pump speed as a percentage of the maximum speed

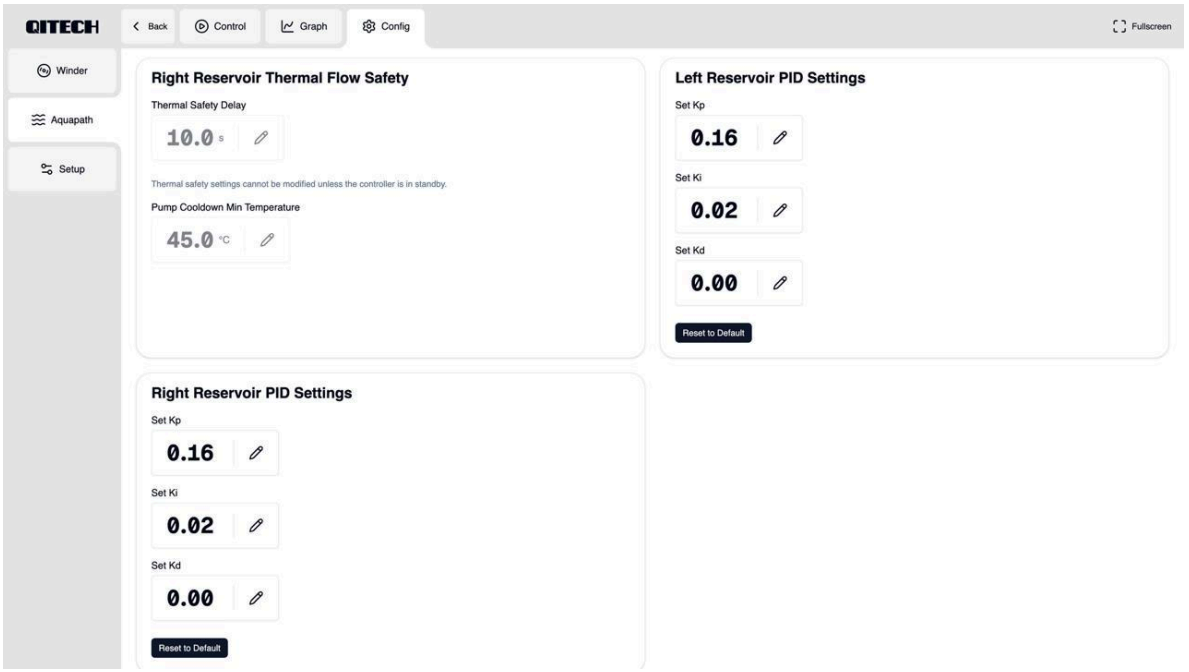
Heating Wattage	Displays the current power output of the heating element in W
Pump	Switches the pump of the circuit on or off
Mode	
Standby	Switches off the pumps, heating and cooling elements, the machine remains switched on
Auto	Activates automatic operation; the pumps run and both circuits are regulated to their set target temperatures
Heating Power Consumption	
Total Power	Displays the current total power consumption of the heating elements in W
Total Energy Consumption	Displays the cumulative energy consumption of the heating elements in kWh



Graph	
Left / Right Reservoir Flow	Shows the water flow rate of the left / right circuit over time
Left / Right Reservoir Temperature	Shows the water temperature of the left / right circuit over time
Heating Wattage	Shows the power output of the heating elements over time
Heating Energy	Shows the energy consumption of the heating elements over time

The graphs can be displayed either as a historical view or as a live view that updates continuously. A time range of up to one hour can be selected. Using the *Add Marker* and *Remove Marker* buttons, specific points in time can be marked manually, named and displayed in the graph. The *Export* button allows the process data to be exported in Excel format.





Config	
Left / Right Reservoir Fan Revolutions	
Set Max Revolution Speed	Sets the maximum speed of the fan of the cooling element as a percentage
Left / Right Reservoir Temperature Tolerances	
Set Heating Tolerance	Sets the permissible deviation from the target temperature before the heating is activated
Set Cooling Tolerance	Sets the permissible deviation from the target temperature before the cooling is activated
Ambient Calibration	
Set Ambient Temperature	Sets the ambient temperature used as a reference by the control system. The "Use Current Sensor Temp" button applies the temperature currently measured by the sensor
Left / Right Reservoir Thermal Flow Safety	
Thermal Safety Delay	Sets the delay after which the heating is switched off if no water flow is detected
Pump Cooldown Min Temperature	Sets the temperature below which the pump switches off after the heating has been deactivated

Left / Right Reservoir PID Settings	
Set Kp / Set Ki / Set Kd	Sets the proportional, integral and derivative parameters of the temperature control loop
Reset to Default	Resets the values of the respective section to their default settings

1.4 Intended use

The QiTech Aquapath is designed for the controlled cooling of extruded filaments with a diameter of 1–3 mm.

The QiTech Aquapath is intended to be operated as part of a process line together with the QiTech Extruder and the QiTech Winder. The machines are connected to one another by the power supply cables. The mains supply and the main switch for the entire line are located on the QiTech Extruder.

The machine must be placed on a stable, level surface with all levelling feet adapted to the groundlevel. The filament is guided continuously through two water reservoirs, which cool it in stages, and then passes a blow-off device that removes residual water from its surface.

The system consists of two separate water circuits, each with its own pump, heating element, cooling element and tank. The pump conveys the water from the tank into the reservoir, from where it flows back through a drain at the bottom of the reservoir. Each circuit can be set to its own temperature. Only distilled water is allowed to use. The blow-off device is operated with compressed air, which must be provided on site by the operator.

The following conditions of use are not permitted:

- Operation in potentially explosive atmospheres
- Operation in food production environments
- Operation outdoors
- Any use beyond the scope defined in these operating instructions

In the event of significant modifications of the machine or deviations from the intended use, a new risk assessment must be carried out and all changes must be documented accordingly.

1.5 Reasonably foreseeable misuse

The following types of use are considered reasonably foreseeable misuse and are not permitted:

Cooling of unsuitable filaments:

- Filaments outside the permitted diameter range of 1–3 mm, which cannot be guided reliably through the guide coins of the system
- Filaments entering the system at a surface temperature above the permissible infeed temperature
- Water-soluble filaments, or filaments that release additives or other substances into the water
- Highly fluid absorbent filaments
- Flat, sharp-edged or pointed filaments, or filaments that are difficult to guide or easily tearable

Improper operation:

- Operating the system outside the intended process line
- Operating the system on an uneven surface or in any position other than the intended horizontal position
- Operating the system with the levelling feet not adapted on the groundlevel
- Operating the system with an empty or insufficiently filled tank
- Operating the system with any medium other than distilled water
- Use of spare parts or components not authorised by the manufacturer
- Operating the system unsupervised or outside of staffed working hours
- Removal or bypassing of safety guards or manipulation of sensors

1.6 Significant modification / other use

Any structural, mechanical or electrical modification to the machine that has not been expressly authorised by QiTech Industries is considered a significant modification. Significant modifications may affect the safety of the machine, render the existing risk assessment invalid and void the CE conformity of the machine. In such cases, a new conformity assessment must be carried out before the machine is put back into service.

Any use of the machine beyond the scope defined in section 1.4 is considered improper use. QiTech Industries and its authorised representatives accept no liability for damage to persons or property resulting from improper use or unauthorised modifications. In such cases, liability rests solely with the operator.

2. Safety

2.1 General safety instructions

These safety instructions apply to all persons working on or with the QiTech Airpath. All personnel must be familiar with these instructions and with the information in the following chapters.

Personnel and authorisation

The machine may only be operated by professionally trained and instructed persons. Uninstructed persons and children must not have access to the machine. Repairs and maintenance work may only be carried out by qualified specialist personnel who are familiar with the relevant occupational safety regulations. Work on the electrical equipment of the machine may only be carried out by a qualified electrician.

Personal protective equipment

Appropriate personal protective equipment must be worn at all times. In particular:

- Always wear slip-resistant safety footwear
- Wear close-fitting clothing; use a hairnet if you have long hair
- Do not wear loose clothing, ties, open shirt sleeves or jewellery
- Hearing protection must be worn if the emission sound pressure level at the workplace exceeds 80 dB(A)

Electrical safety

Before carrying out any maintenance, cleaning or repair work, the machine must be switched off and fully de-energised. The following steps must be observed:

- Switch off the process line at the main switch on the QiTech Extruder
- Disconnect the mains plug to secure the machine against unintentional restart
- Disconnect the LAN cable and compressed air connection
- Cover or isolate neighbouring live parts

The mains supply cable may only be connected to a socket outlet with earthing contact or a suitable connection box. The machine must be connected to a mains circuit protected by a residual current device (RCD). Before connection, have the socket outlet or connection box checked by a qualified electrician. Never use the machine with defective, crushed or bare cables.

If water escapes from the machine, switch it off and disconnect the mains plug immediately. Do not touch or continue to operate the machine until the cause of the leak has been identified and rectified and the absence of voltage has been confirmed.

General operational safety

Before starting work, carry out a visual inspection to ensure the machine is in perfect technical condition. The machine may only be used when all protective devices are properly fitted and fully operational. Never leave the machine in operation without supervision.

- Keep the work area clean and free of obstacles; do not place objects on the machine
- Switch off the machine when it is not in use
- Protect the electrical components from moisture
- Check the machine regularly for damage; only use original spare parts authorised by the manufacturer
- Ensure adequate lighting in accordance with applicable national workplace regulations

The water in the heated reservoirs can reach high temperatures. Do not reach into the reservoirs and avoid contact with the hot water during operation.

QiTech Industries accepts no liability for damage caused by failure to observe these operating instructions. The use of non-authorised spare parts or accessories will invalidate the guarantee and warranty.



Danger to life due to technically faulty machine
The machine may only be used when it is in perfect technical condition, in accordance with its intended use and in compliance with these operating instructions. All faults, especially those that could impair safety, must be rectified immediately before starting the machine.

2.2 Pictograms, symbols and warning notices

This chapter provides a comprehensive reference of all safety symbols that operators may encounter in an industrial working environment. Graphical symbols, including warning signs, mandatory signs, prohibition signs, follow ISO 7010. All people working with the QiTech Aquapath must be familiar with the meanings of these symbols.

The general warning symbol can be used in combination with one of the following signal words:

DANGER Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.

WARNING Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

NOTICE Indicates important information not related to personal injury. Non-observance may result in damage to property or the environment.

Warning signs		Mandatory signs	
	W001 General warning sign		M001 General warning sign
	W004 Warning of laser beam		M003 Use hearing protection
	W011 Warning of slippery surface		M004 Use eye protection
	W012 Warning of electrical voltage		M008 Use foot protection
	W016 Warning of toxic substances		M009 Use hand protection
	W017 Warning of hot surface		M010 Use protective clothing
	W018 Warning against automatic start-up		M012 Use handrail

	W019 Warning of crushing hazard		M014 Use head protection
	W024 Warning against hand injuries		M021 Switch off before maintenance or repair
	W025 Warning against counter-rotating rollers		
Prohibition signs		**Other pictograms**	
	P001 General prohibition sign		F001 Fire extinguisher
	P002 No smoking		WEEE symbol Separate collection of waste electrical and electronic equipment
	P003 No open flames; fire, open source of ignition and smoking not permitted		
	P022 Eating and drinking not permitted		

	P024 Entering the area not permitted		
	P031 Switching not permitted		
	D-P006 Access not permitted for unauthorised persons		

2.3 Safety devices

The QiTech Aquapath is equipped with the following safety devices. All safety devices must be fully operational before the machine is started. It is not permitted to operate the machine with any safety device removed, bypassed or disabled. If a safety device is opened or removed for maintenance purposes, it must be correctly refitted and verified before the machine is switched on again.

Flow monitoring
One flow sensor per reservoir monitors the water flow through the pipes. If no water flows through the pipes, the heating and cooling switch off automatically. This prevents the heating and cooling elements from operating without water.

Overheating protection
The heating is equipped with an overheating protection. If the heating exceeds its permissible operating temperature, it switches off automatically. The heating should not be restarted until it has cooled sufficiently and the cause of overheating has been identified.

Notice: If safety-relevant changes or defects are detected on the machine, shut it down immediately, secure it against restart and report the issue to the responsible person before resuming operation.

2.4 Qualification of personnel / instruction and training

All persons working with the QiTech Aquapath must have the qualifications appropriate to their assigned tasks. The following personnel categories are defined for the purposes of these operating instructions:

Operator: The operator is the person who uses, adjusts and monitors the machine in day-to-day operation. The operator is characterised by practical experience in handling the machine as well as in-depth knowledge of its functions and the correct course of action in the event of a fault.

Instructed person: An instructed person has been briefed by the operator on the tasks assigned to them and on the potential hazards associated with improper behaviour. The correct execution of all activities carried out by an instructed person is monitored by specialised personnel.

Specialised personnel: Specialised personnel are able to carry out their assigned tasks independently and to recognise and avoid potential hazards, based on their specialist training, knowledge and experience as well as their familiarity with the relevant regulations. It is assumed that specialised personnel have also received the instruction described above.

Qualified electrician: A qualified electrician is a person who, by virtue of their technical training, knowledge and experience as well as their familiarity with the relevant regulations, is able to assess the work assigned to them and recognise and avoid potential hazards related to electrical systems.

Tasks	**Minimum qualification**
Transport	Specialised personnel of the manufacturer
Installation and commissioning	Specialised personnel of the manufacturer
Day-to-day operation	Instructed personnel of the operator
Setting, adjusting and restarting	Specialised personnel of the operator or manufacturer
Troubleshooting and fault rectification	Specialised personnel of the operator or manufacturer
Cleaning and maintenance	Specialised personnel of the operator or manufacturer
Decommissioning	Specialised personnel of the operator

As the operator, you are obliged to ensure that all personnel working on or with the machine are trained and instructed in accordance with these operating instructions before starting work. This includes instruction on all applicable legal and accident prevention regulations as well as on all safety devices on and around the machine. The varying technical qualifications of individual employees must be taken into account when planning instruction sessions.

Instruction must be understood and retained by all personnel. Compliance with instruction content must be verified regularly. Written confirmation of attendance must be obtained from each employee following each instruction session.

Restrictions on employment

The following persons must not work on or with the machine under any circumstances:

- Persons with skin diseases
- Persons with chronic respiratory conditions or a tendency to allergic reactions
- Pregnant women and nursing mothers
- Persons under 18 years of age

2.5 Residual risks

Despite the safety measures integrated into the design and construction of the QiTech Aquapath, certain residual risks cannot be fully eliminated. The following hazards remain and must be taken into account by all persons working on or with the machine.



Danger! Operation only permitted with all protective devices fitted
The machine must only be operated with all protective devices correctly installed and fully functional. The operator is responsible for ensuring this before each use.



Danger! Risk of scalding from hot water
The water in the heated reservoirs can reach temperatures of up to 80 °C. Do not reach into the reservoirs and avoid all contact with the hot water during operation. Allow the water to cool before emptying the reservoirs or the tank.



Danger! Slipping on wet floors
Water on the floor around the machine creates a slipping hazard. Wear slip-resistant footwear and remove spilled water immediately.



Danger! Electric shock
Before carrying out any repair, set-up or maintenance work, the machine must be fully de-energised. Do not resume work until the absence of voltage has been confirmed.



Danger! Electric shock from water contact with electrical parts
If water comes into contact with electrical components, there is a risk of electric shock. Switch off the process line and disconnect the mains plug immediately if a leak occurs, and do not resume operation until the cause has been rectified.

3. Transport, installation and commissioning

3.1 Transport

Warning! Risk of injury due to improper transport
When loading and unloading the machine, do not place any limbs under machine parts when setting them down. Do not stand under suspended loads or between moving loads and walls or other machine parts. Wear appropriate personal protective equipment during all transport operations.

The QiTech Aquapath is delivered fully assembled on its mobile underframe.

Before transport, the following steps must be carried out:

- Drain the water from both tanks before moving the machine
- Switch off the process line at the switch on the QiTech Extruder
- Disconnect and secure all cables and accessories to prevent damage during transport
- Raise all levelling feet before moving the machine

The machine may be moved manually on its castors over short distances on a level surface. For longer distances or loading onto a vehicle, suitable lifting equipment must be used. The permissible load-bearing capacity of the lifting equipment must meet the machine weight. After transport, inspect the machine for any damage that may have occurred during transit.

Caution! Risk of damage due to incorrect installation and commissioning
Errors during installation and commissioning can cause considerable damage to property. Installation and commissioning may only be carried out by the manufacturer's employees.

3.2 Requirements for the installation site

Before installing the machine, ensure that the installation site meets the following requirements:

- The surface must be level, solid and capable of supporting the weight of the machine
- The location must be used exclusively for industrial or commercial purposes
- The location must be as fire-free as possible and equipped with a fire extinguisher
- The floor and the immediate surroundings must be resistant to moisture
- Sufficient clearance must be maintained around the machine to allow safe operation and unobstructed access for maintenance work
- Adequate lighting must be provided in accordance with applicable national workplace regulations

3.3 Basic installation

Caution! Risk of damage due to incorrect installation and commissioning
Errors during installation and commissioning can cause considerable damage to property. Installation and commissioning may only be carried out by the manufacturer's employees.

The QiTech Aquapath is delivered fully assembled on its mobile underframe. After positioning the machine at the intended installation site, carry out the following steps:

- Position the machine at the intended installation site in accordance with the requirements
- Adapt all levelling feet to secure the machine against unintentional movement
- Connect the power supply cable from the QiTech Extruder to the Aquapath, and the power supply cable from the Aquapath to the QiTech Winder
- Connect the LAN cables: the "Out" port of each machine is connected to the "In" port of the next machine. The ports are labelled accordingly.
- Route all cables in a way that they cannot be tripped over
- Connect the machine to the compressed air supply provided on site

The machine is delivered with the latest available version of the QiTech Control software installed.

3.4 Electrical connection

Danger! Electric shock due to improper electrical connection
All work on the electrical connection must be carried out by a qualified electrician. Ensure the mains socket is de-energised before connecting the machine.

Before connecting the machine to the mains supply, carry out the following steps:

- Verify that the mains voltage and frequency at the installation site correspond to the specification of the machine
- Ensure that the mains socket is equipped with an earthing contact and has been checked by a qualified electrician
- Ensure that the mains circuit is protected by a residual current device (RCD)
- Only use original fuses with the prescribed current ratings to avoid overloading

Ensure that proper earthing of the electrical system is maintained through protective conductor systems at all times.

3.5 Commissioning

Before switching on the machine for the first time, carry out a full visual inspection:

- All protective devices are correctly fitted
- All cables are undamaged and correctly routed
- All levelling feet are adapted to the ground level

- Both tanks are filled with distilled water and no leaks are present
- The compressed air supply is connected

Once all checks have been completed satisfactorily, the machine may be switched on.

4. Operation

4.1 Filling and emptying the water tank

Filling the tanks

To fill the tanks, open the maintenance door and fill each of the two tanks with 16 l of distilled water. The tanks have a volume of 20 l, but should be filled by 80 %. Only distilled water may be used, to avoid contamination with bacteria.

Emptying the tanks

When the machine is switched off, the pumps stop and the water flows back from the reservoirs into the tanks. The reservoirs therefore need to be emptied separately. To empty the tanks, the hose is disconnected downstream of the pump and directed into a separate container or a drain, so that the pump conveys the water out of the system instead of into the reservoir.

- Allow the water to cool before draining
- Switch off the pump in the QiTech Control software
- Close the shut-off valve
- Disconnect the hose downstream of the valve and place it into a separate container
- Open the shut-off valve and switch on the pump to convey the water out of the tank
- Switch off the pump once the tank is empty, close the valve and reconnect the hose

If the machine is not to be used for a prolonged period, the tanks must be emptied. This prevents standing water from remaining in the system and germs from forming.

The water must be disposed of in accordance with applicable national regulations, as it may has come into contact with critical substances.

4.2 Starting the machine

A visual inspection must be carried out every working day before the machine is started. Verify that all protective devices are correctly fitted, all cables are undamaged, and the machine shows no signs of leakage.

- Ensure that both tanks are filled with distilled water
- Switch on the process line at the main switch on the QiTech Extruder
- Ensure that the compressed air supply for the blow-off device is connected
- The Aquapath will appear in the left column of the QiTech Control software

Once the machine is switched on, the pumps convey the water from the tanks into the reservoirs, from where it flows back into the tanks.

4.3 Cooling the filament

The QiTech Aquapath cools the filament in two stages. Each reservoir can be set to its own temperature. It is recommended to operate the first reservoir at a higher temperature and the second reservoir at a lower temperature, so that the filament is cooled gradually.

Setting the reservoir temperatures

- Select "Control" in the function bar of the QiTech Control software
- Set the target temperature for each reservoir
- The heating and cooling elements maintain the water at the set temperature

The target temperature should be reached before the filament is fed in. The machine does not detect whether filament is present in the reservoirs.

Feeding in the filament

The filament is pulled through the system by the puller of the QiTech Winder.

- Guide the filament by hand through both reservoirs and through the blow-off device, using a tool such as pliers
- Close the blow-off device onto the filament
- Insert the three guide coins by hand: at the beginning and end of the first reservoir, and at the beginning of the second reservoir. The blow-off device replaces the guide coin at the end of the second reservoir

Danger! Risk of scalding from hot water
The water in the reservoirs can reach temperatures of up to 80 °C. Avoid contact with the hot water. When the hot filament enters the water, splashes or steam may be produced.

4.4 Stopping and switching off

To shut down the machine:

- Select "Standby" mode in the QiTech Control software – the pumps and the heating and cooling elements are switched off
- Wait until the water has flowed back from the reservoirs into the tanks
- Switch off the process line at the main switch on the QiTech Extruder

If the machine is not to be used for a prolonged period, empty the tanks as described in section 4.1 and disconnect the mains plug.

Notice: If safety-relevant changes or defects are detected during operation, shut down the machine immediately with the main switch on the QiTech Extruder and report the issue to the responsible person before resuming operation.

5. Maintenance and inspection

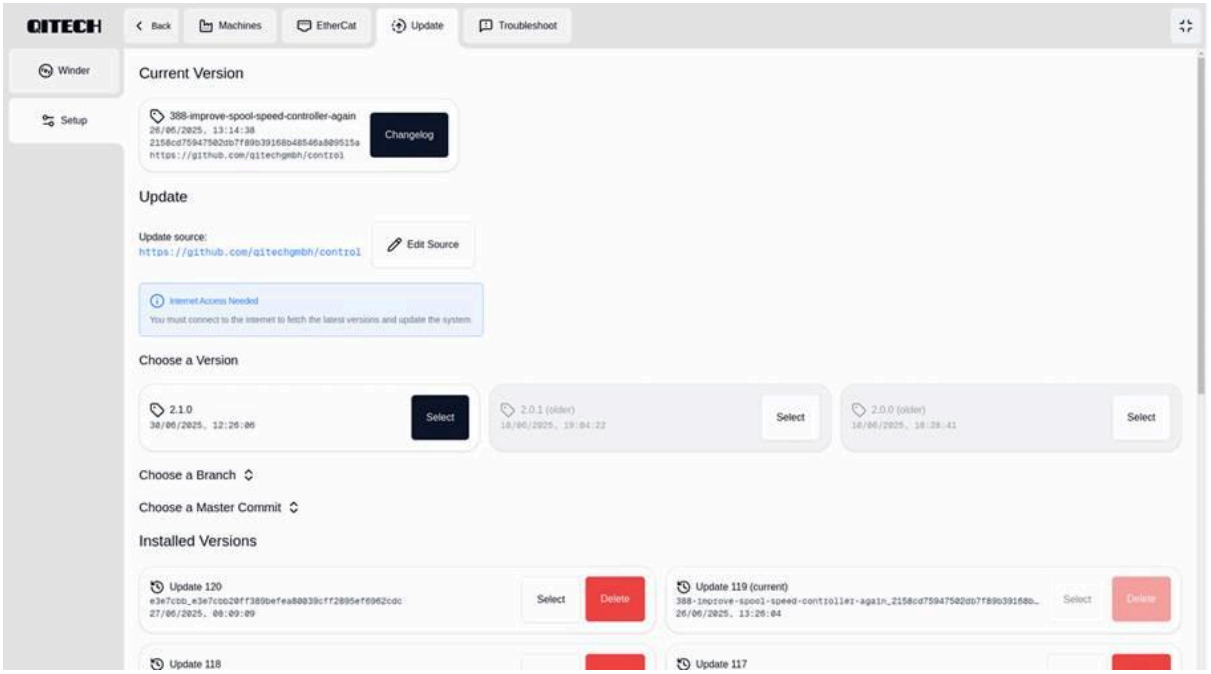
5.1 Updating the QiTech Control software

Software updates may contain bug fixes, new functions or security improvements. It is therefore recommended to check regularly whether updates are available and to install them promptly.

To perform a software update, proceed as follows:

- Ensure the machine has an active internet connection
- Open the QiTech Control software and navigate to *Setup* and *Updates*
- Available updates will be displayed; select the desired update with the *Select* button
- The display will shut down and restart automatically; the updated software version will be installed upon restart

Previous software versions and the complete update history can also be viewed under *Setup* and *Updates*.



Notice: Modifying the software source or installing unofficial software versions may compromise the safety and functionality of the machine and will invalidate the guarantee and warranty. Only official updates from QiTech Automatics GmbH may be installed.

5.2 Mechanical maintenance

Checking the blow-off device and guide coins

The blow-off device and all filament guide coins should be checked regularly for wear and contamination. Replace them if necessary. Proceed as follows:

- Switch off the process line and disconnect the mains plug to secure the machine against unintentional restart
- Check the blow-off device and the guide coins for visible wear or damage
- Replace any worn guide coins with new parts

Checking for leaks and retightening screw connections

To prevent water from escaping, the machine should be checked regularly for leaks, and all screw connections should be retightened. Proceed as follows:

- Switch off the process line and disconnect the mains plug to secure the machine against unintentional restart
- Check all water-carrying components, connections and reservoirs for leaks
- Retighten all screw connections

5.3 Water maintenance

Changing the water

The water must be changed if it shows visible contamination or if the QiTech Aquapath is not used for a prolonged time. Standing water that has been heated promotes the growth of germs; regular water changes are necessary for hygienic operation. Proceed as follows:

- Switch off the machine and disconnect the mains plug
- Allow the water to cool before draining
- Drain the water from both tanks
- Refill the tanks with distilled water

Cleaning the water filter

The water filters must be cleaned with distilled water if the water shows visible contamination or if the QiTech Aquapath is not used for a prolonged time. Proceed as follows:

- Switch off the process line and disconnect the mains plug to secure the machine against unintentional restart
- Remove both water filters
- Clean the filters with distilled water
- Refit both filters

6. Troubleshooting

The following table provides an overview of the most common faults, their possible causes and recommended corrective actions. If a fault cannot be resolved using the measures described below, contact QiTech Industries GmbH for technical support.

Before attempting to rectify any fault, ensure that the machine is in a safe condition. For faults requiring access to internal components, switch off the machine and disconnect the main plug.

Fault	Possible cause	Corrective action
Machine is not recognised by Control software	LAN cable not connected	Check and reconnect the LAN cable
	Outdated software version	Update the QiTech Control software
	Network configuration error	Check network settings
Pump does not convey water	Air in the pump	Open the shut-off valve until water flows out, so that the air can escape
	Tank empty or filled below the minimum level	Fill the tank with distilled water
	Shut-off valve closed	Open the shut-off valve
Water temperature is not adapted to the target temperature	Cooling element not working	Check whether the fan of the cooling element is running
	Heating switched off by the flow monitoring	Check the water flow; the heating switches off automatically if no water flows through the circuit

	Overheating protection triggered	Allow the heating to cool and identify the cause of overheating
Filament is not dry after the blow-off device	Compressed air supply not connected	Connect the compressed air supply
	Air pressure too low	Increase the air pressure, maximum 8 bar
Water is contaminated	Water not changed for a long time	Change the water and clean the water filter
	Unsuitable filament material used	Verify that the filament does not release substances into the water
Machine does not start	Emergency stop on the QiTech Extruder switch is actuated	Release the emergency stop switch by rotating it clockwise, then restart the machine
	Mains plug not connected	Check the mains connection and the power supply

7. Decommissioning, disassembly and disposal

When the machine has reached the end of its planned service life, QiTech Industries GmbH recommends contacting the manufacturer to arrange for the machine to be taken back. If this is not possible, the machine must be permanently taken out of service and disposed in accordance with the instructions in this chapter and in compliance with applicable national regulations.

Disassembly and decommissioning may only be carried out by trained and qualified specialist personnel of the operator.

Decommissioning

Before disassembly begins, the machine must be taken out of service by properly switching off, disconnect the power supply and secure the machine against unintentional restart. Allow the machine to cool completely before proceeding with disassembly. Drain all water from the resrvoirs and the tank, then leave the machine until residual moisture is dried.

Disassembly

The following tools are required for disassembly:

- M5 Allen key
- Standard screwdriver set

The following personal protective equipment must be worn during disassembly:

- Safety footwear
- Protective gloves
- Protective clothing

The machine consists largely of aluminium and plastic components, which can be easily separated by loosening the screw connections. Proceed as follows:

- Remove all cables and set aside for separate disposal
- Remove the aluminium composite panels from the housing frame by loosening the screw connections
- Separate the mechanical components from the electronic components
- Remove and separately collect all auxiliary materials

Disposal

All components must be separated by material class to enable subsequent recycling or differentiated scrapping. The following materials require separate disposal by an authorised specialist company:

- Electrical and electronic components
- Lubricants and oils

Equipment bearing the WEEE symbol must not be disposed of as unsorted waste. The operator is responsible for ensuring that all waste is separated by material and disposed of via an authorised collection or take-back point in accordance with applicable national regulations.

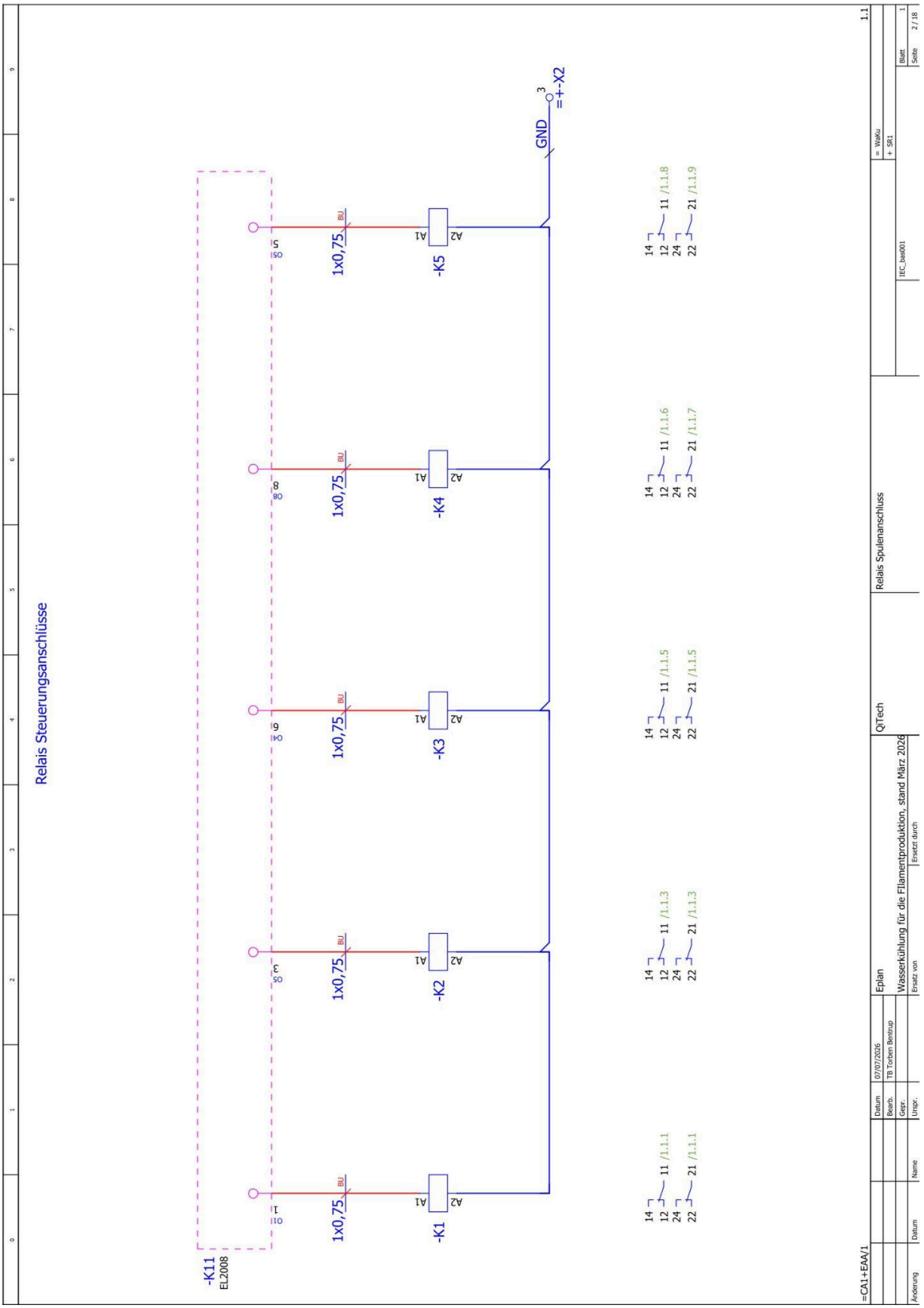
Notice: Disposal of hazardous materials such as lubricants, oils and electronic components must be carried out in accordance with applicable national environmental regulations. Contact an authorised waste disposal specialist if in doubt.

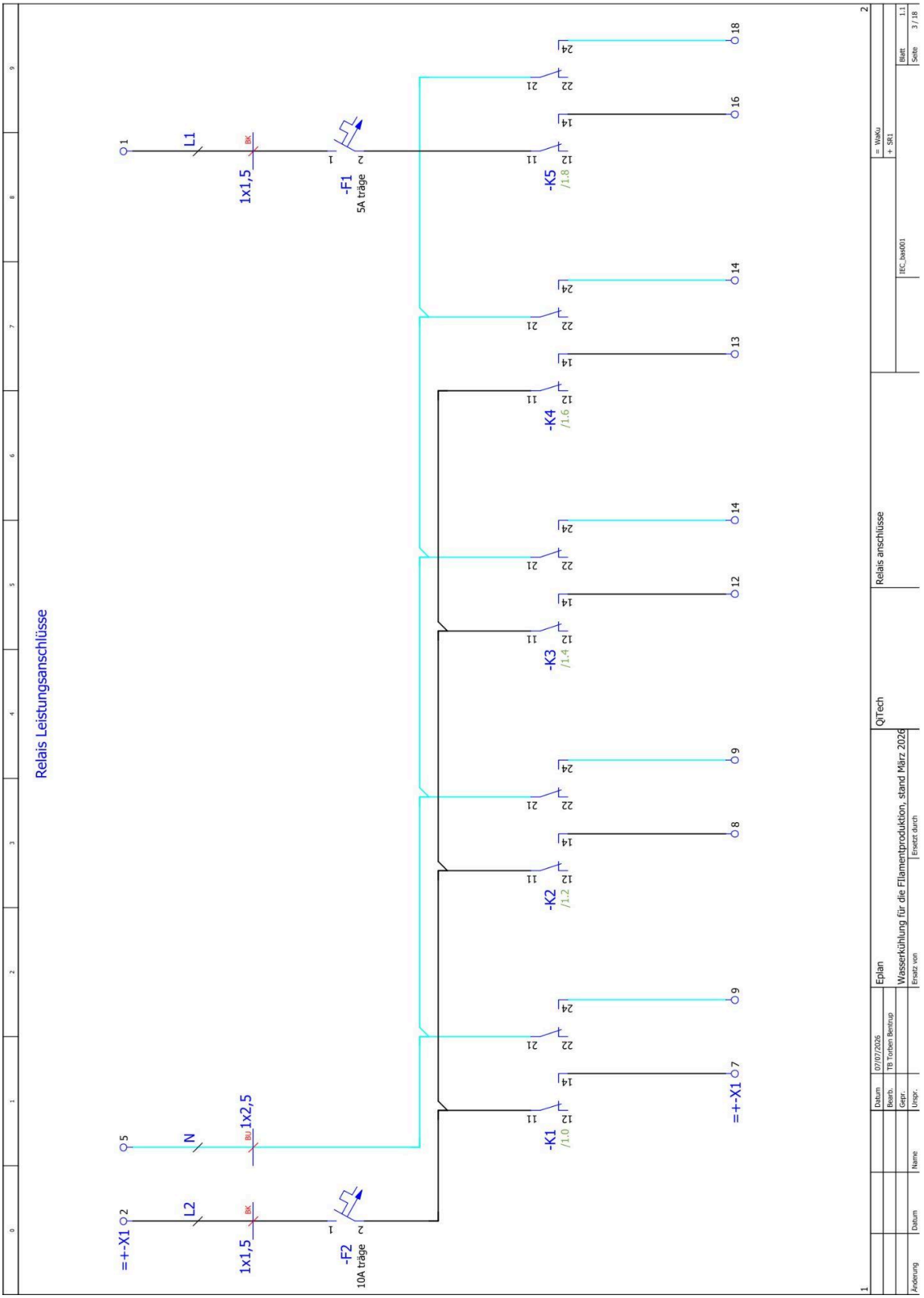
8. Annexes

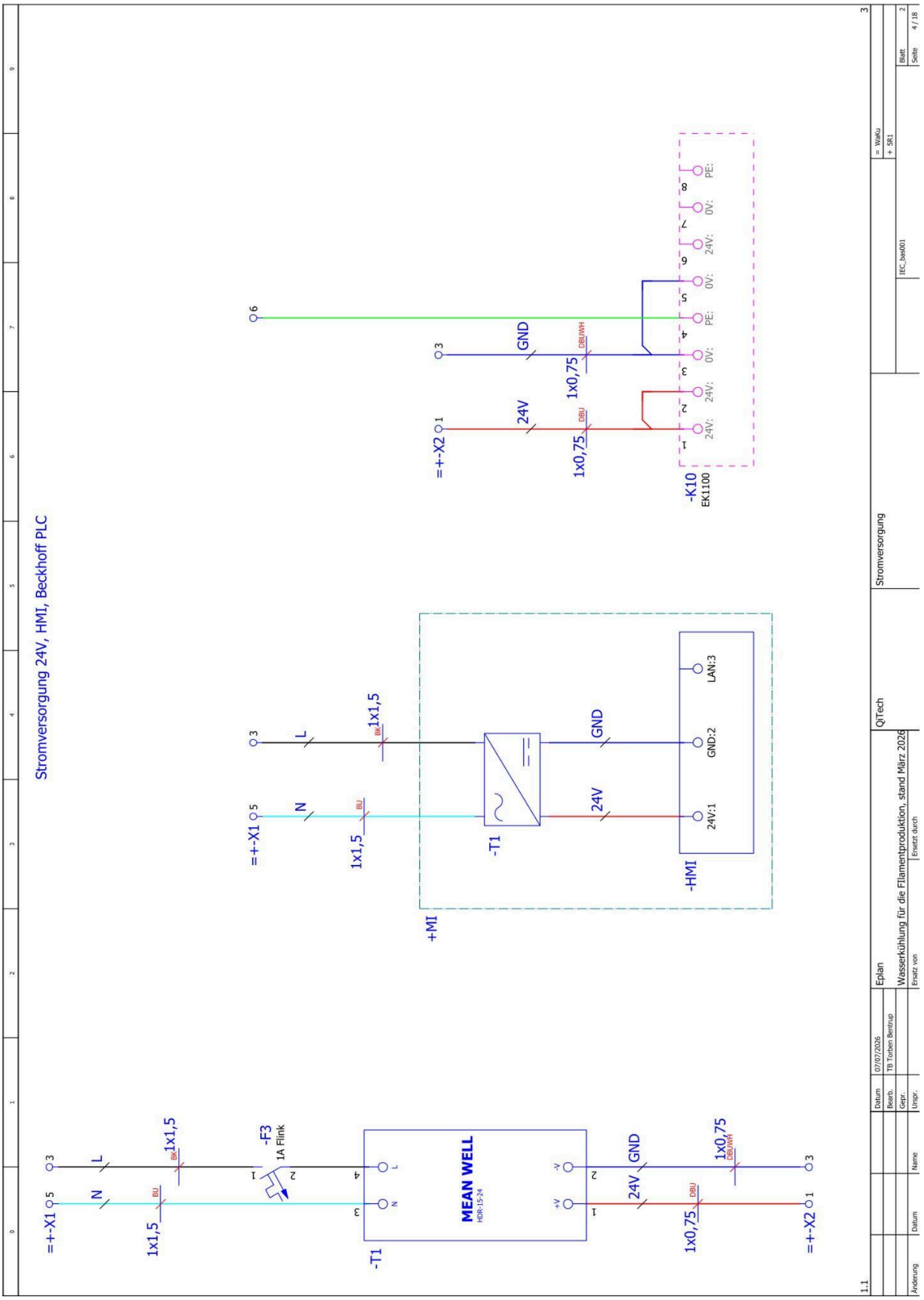
Annex A: Rendering of the process line

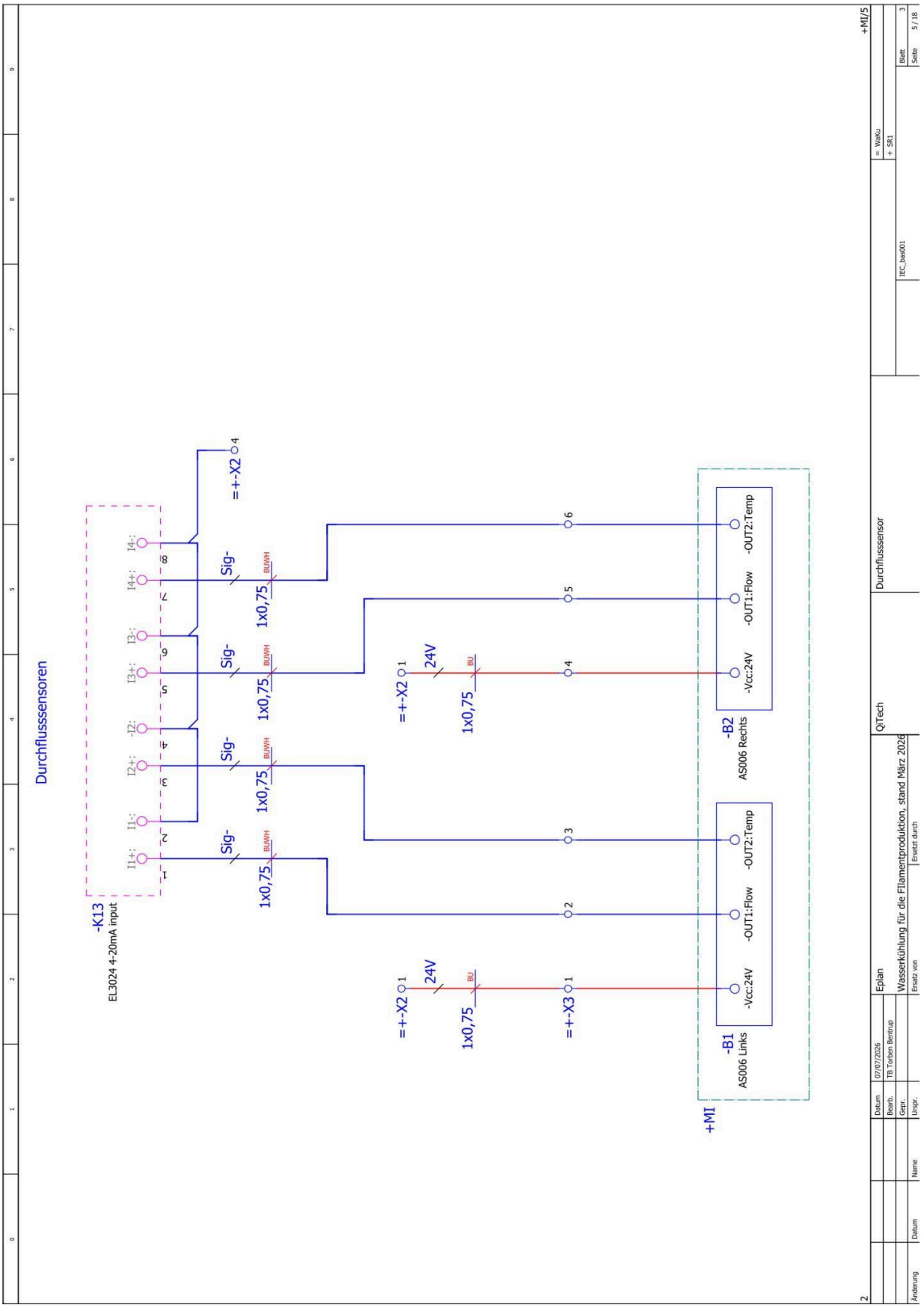


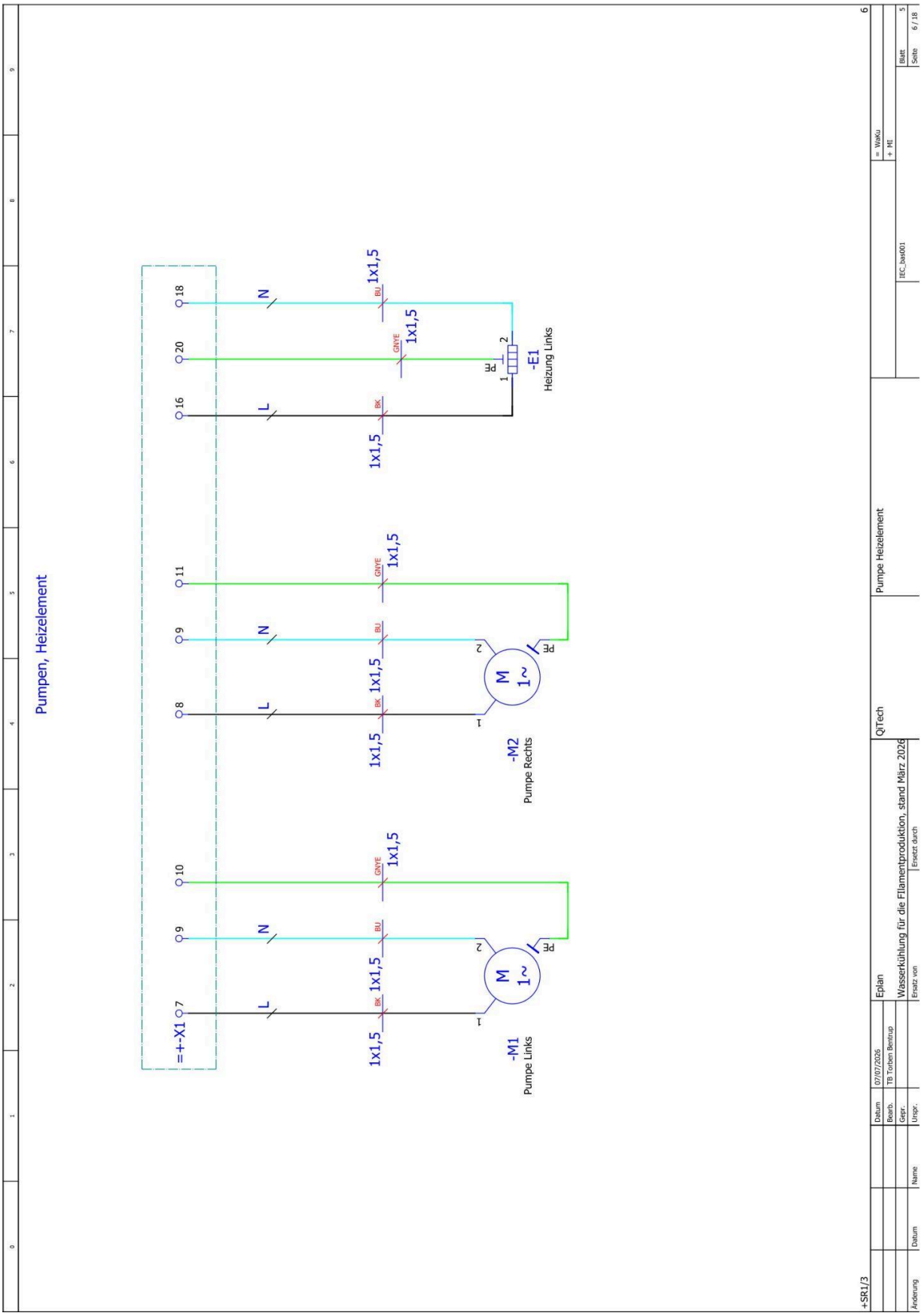
Annex B: Electrical documentation

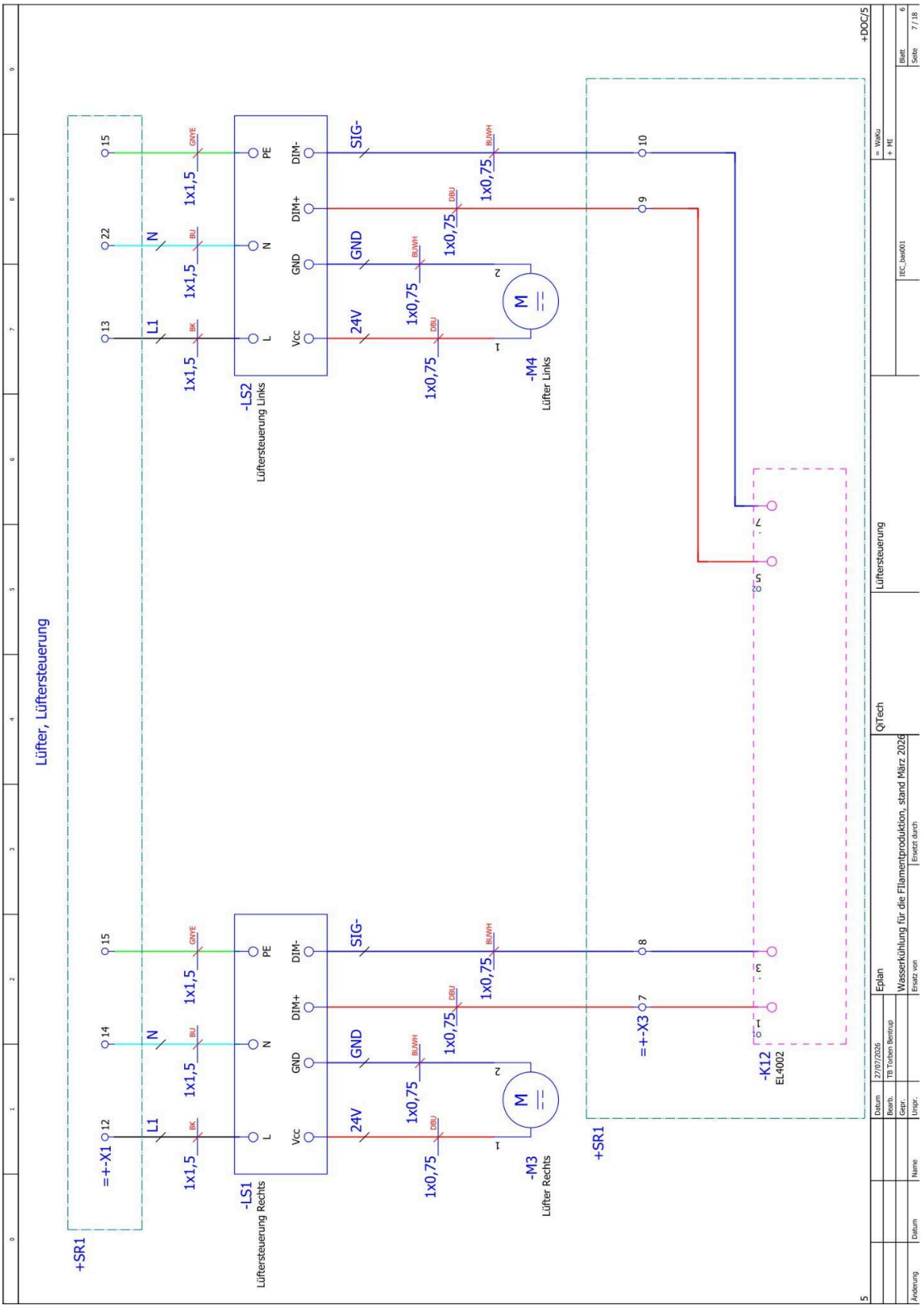



Relais Leistungsanschlüsse
=+-X1
L1
L2
N
1x1,5
1x2,5
-F1
5A träge
-F2
10A träge
-K1
-K2
-K3
-K4
-K5
=+-X1
Relais anschlüsse
QiTech
Eplan
WasserKühlung für die Filamentproduktion, stand März 2026
07/07/2026
TB Torben Bentrup


Stromversorgung 24V, HMI, Beckhoff PLC
=+-X1
=+-X2
-F3
1A Flink
-T1
MEAN WELL
HDR-15-24
-K10
EK1100
+MI
-HMI
-T1
GND
24V
1x1,5
1x0,75
QiTech
Eplan
Stromversorgung
Wasserkühlung für die Filamentproduktion, stand März 2026


Durchflusssensoren
-K13
EL3024 4-20mA input
I1+:
I1-:
I2+:
I2-:
I3+:
I3-:
I4+:
I4-:
Sig-
1x0,75
BUWH
=+-X2
=+-X3
24V
BU
+MI
-B1
AS006 Links
-B2
AS006 Rechts
-Vcc:24V
-OUT1:Flow
-OUT2:Temp
+MI/5
Durchflusssensor
QiTech
Eplan
Wasserkühlung für die Filamentproduktion, stand März 2026
07/07/2026
TB Torben Bentrup
Datum
Bearb.
Gepr.
Urspr.
Ersatz von
Ersetzt durch
Änderung
Name
= WaKu
+ SR1
IEC_basic001
Blatt 3
Seite 5 / 18


Pumpen, Heizelement
=+-X1
-M1
Pumpe Links
-M2
Pumpe Rechts
-E1
Heizung Links
Pumpe Heizelement
Wasserkühlung für die Filamentproduktion, stand März 2026
Eplan
QiTech
+SR1/3


Lüfter, Lüftersteuerung
+SR1
=+-X1
-LS1
Lüftersteuerung Rechts
-LS2
Lüftersteuerung Links
-M3
Lüfter Rechts
-M4
Lüfter Links
-K12
EL4002
=+-X3
+SR1
L1
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PE
Vcc
GND
24V
SIG-
DIM+
DIM-
1x1,5
1x0,75
GNYE
BU
BK
BUWH
DBU
+DOC/5
QiTech
Eplan
Lüftersteuerung
Wasserkühlung für die Filamentproduktion, stand März 2026
27/07/2026
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